

Math Camp: Day 1

Yale University

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Variables and Constants

- Constant: A concept or measure that has a single value for a given set.
- Variable: A concept or measure that takes different values in a given set.
- Constant or Variable?
 - Number of ethnic groups in a country for all countries in the world
 - Highest threshold of per-capita income of a country, in which democracy has collapsed

Sets

Sets are essentially collections of elements.

- A set is denoted as $A = \{a, b, c\}$, which means set A contains the **elements** a, b, c .
- Sets can numbers as elements, but they can also contain other types of elements: strategies two competing firms might use.

Common Sets:

- $\mathbb{N}$ Natural Numbers
- $\mathbb{Z}$ Integers
- $\mathbb{Q}$ Rational Numbers
- $\mathbb{R}$ Real Numbers
- $\mathbb{C}$ Complex Numbers
- Subscript: $\mathbb{Z}_+$ Positive (Negative) Values of a Set
- Superscript: $\mathbb{Z}^d$ Dimensionality (number of dimensions)

Properties of Sets

- The number of elements in a set can be **finite** or **infinite**.
 - What are examples of finite and infinite sets?
- If the elements in a set can be counted the set is **countable**. If not, the set is **uncountable**
- A **bounded** set has a finite size, but an **unbounded** set does not.
 - What is the difference between a bounded set and a finite set?
 - A bounded set can have an infinite number of elements.

Properties of Sets

- The notation $a \in A$ denotes that a is an element of set A . The notation $b \notin B$ denotes a is not an element of B .
- The notation $A \subset X$ denotes that A is a proper subset of X . This means that all of the elements in A and that X has at least one additional element.
- The notation $A \subseteq X$ denotes that A is a subset of X . This means that all of the elements in A but A and X could have the same elements.
- The **Cardinality of a set** is the number of elements in the set.
- A **Singleton** is a set with only one element
- $\{\}$ or $\emptyset$ denotes an empty set or a set with no elements.
- The **universal set** contains all elements
- In an **ordered set** the order matters, so sets with the same elements that are listed in different orders are distinct. Sets can also be **unordered sets**.

Operators

Operators: operate of sets and elements of sets

- What are some examples of operators?

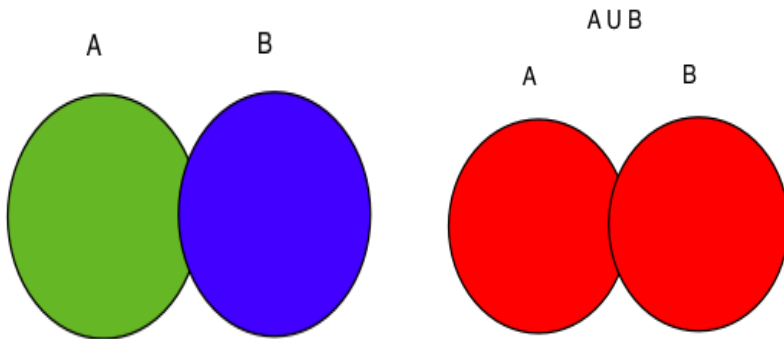
Set Operators

- $A \cup B$ denotes the **union** between the sets A and B . $A \cup B$ is the set of elements that are contained in *either* A or B .
 - If $A = \{1, 2, 3\}$ and $B = \{2, 3, 4\}$ what is $A \cup B$?
- $A \cap B$ denotes the **intersection** between A and B . $A \cap B$ is the set of elements that are contained in *both* A and B .
 - If $A = \{1, 2, 3\}$ and $B = \{2, 3, 4\}$ what is $A \cap B$?
- If $A \subset X$, then A^C denotes the **complement** of A . A^C is the set of elements that are contained in X but not in A or $A^C = \{x \in X \text{ such that } x \notin A\}$
- The **difference** between two sets A and B , denoted $A \setminus B$ is the set containing all of the elements of A that are not in B .
 - If $A = \{1, 2, 3\}$ and $B = \{2, 3, 4\}$ what is $A \setminus B$?

Set Operators Continued

- **Mutually exclusive** sets are sets with an intersection equal to an empty set.
- A **partition** is a collection of subsets whose union forms the set.
- $A \times B$ denotes the **Cartesian Product** of the sets A and B . This is the set of all possible ordered pairs whose first component is an element of A and whose second component is an element of B or $A \times B = \{(a, b) \text{ such that } a \in A \text{ and } b \in B\}$
 - Consider the sets $A = \{\text{Civil War, Peace}\}$ and $B = \{\text{Democracy, Non-Democracy}\}$. What is the Cartesian product of these sets?

Visualizing Sets



Levels of Measurement and Relations

What is a a relation? What is an example of a binary relation

Level of Measurement	Mathematical Relation	Meaningful Distance	Zero is Meaningful
Nominal			
Ordinal	x		
Interval	x	x	
Ratio	x	x	x

Levels of Measurement Examples

- Nominal
 - Democracy, Gender
- Ordinal
 - Approval of a Candidate: Very Bad, Bad, Good, Very Good
- Interval
 - Approval of a Candidate: 0-10 scale
- Ratio
 - Provincial Budget

Other Important Operators

- Summation Operator: $\sum_{i=k}^l x_i$ “The sum of x_i over the range from $i = k$ through $i = l$.”
- $\sum_{i=1}^3 x_i = x_1 + x_2 + x_3$
 - Total Spending of Provincial Governments in Mexico: $\sum_{p=1}^{31} B_p$
- Product Operator: $\prod_{i=k}^l x_i$
- $\prod_{i=1}^3 x_i = x_1 * x_2 * x_3$
- Factorial notation denotes the product of a specific sequence of numbers: $n! = n(n-1)(n-2)\dots(2)(1)$

Basic Notation for Proofs

- $A \implies B$. A implies B .
- $A \impliedby B$. B implies A .
 - What is a another term for $\implies$?
 - If $A \implies B$, does $B \implies A$?
- $A \iff B$. A implies B and B implies A .
 - A holds if and only if (iff) B holds
- $\neg A$. Not A , or negation of A .
 - If $A \implies B$, $\neg B \implies \neg A$. Examples?
- Set of Integers: $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$
- Set of Strictly Positive Integers: $\mathbb{Z}^{++} = \mathbb{N} = \{1, 2, 3, \dots\}$
- Set of Rational Numbers: $\mathbb{Q} = \{x | x = \frac{p}{q}, p, q \in \mathbb{Z}, q \neq 0\}$

Symbols for Proofs

- $\forall$: For all, for any
- $\exists$: There exists
- $\in$: Is contained in, is an element of
- $\ni$: Such that, contains as an element
- $\subset$: Is a subset of
- *QED* Latin for “quod erat demonstrandum” or “which was to be proven”.

Proofs

We seek to prove that $A \implies B$. We will discuss five strategies to prove statements:

- Direct Proofs
 - Deductive Reasoning
 - Contrapositive
- Indirect Proofs
 - Contradiction
 - Induction

Deductive Reasoning

Deductive Reasoning relies on a series of accepted axioms or theorems such that

$A_0 \implies A_1 \implies A_2 \implies \dots \implies A_{n-1} \implies A_n$, where $A = A_0$ and $B = A_n$. The crux of this strategy is to find the intermediate steps.

Example: Prove the number three is an odd number. $3 \implies \text{odd}$

Proof: A number q is odd if $\exists$ an integer m such that $q = 2m + 1$.
Let $m = 1$. Then $2m + 1 = 3$ Therefore, 3 is an odd number.

Contrapositive

For a contrapositive proof we simply use the fact that $A \implies B$ is equivalent to $\neg B \implies \neg A$.

Example: Let x and y be two positive numbers. Prove that if $xy > 9$, then $x > 3$ or $y > 3$.

- $\neg B$: Suppose that both $x \leq 3$ and $y \leq 3$.
- $\neg B \implies \neg A$: If $x \leq 3$ and $y \leq 3$, then $xy \leq 9$

Contradiction

We can prove $A \implies B$, by assuming that $A \nRightarrow B$, and finding a contradiction. That is assume that A is true but B is not true. If this yields a contradiction we conclude that: B must be true or A is false. But we have assumed that A is true and so it must be the case that B is true.

Contradiction: Example

Example Prove that $\sqrt{2}$ is an irrational number.

- $A \nRightarrow B$: $\sqrt{2}$ can be expressed as $\frac{p}{q}$, where p and q are not both even numbers (given that $1 < \sqrt{2} < 2$)
- $\sqrt{2} = \frac{p}{q} \implies 2 = \frac{p^2}{q^2} \implies 2q^2 = p^2 \implies p$ is even
- If p is even, then $\exists$ an integer m such that $2m = p$.
- if $2m = p$, where m is an integer, then $4m^2 = p^2 = 2q^2 \implies q^2 = 2m^2$
- If $q^2 = 2m^2$, then q is an even number, which is a contradiction.

Induction

Induction is used for propositions regarding integers or that are indexed by integers. Consider a list of statements:

$P(1), P(2), \dots, P(n)$. If we can prove *two* statements about the sequence, then every statement in the entire sequence is true.

- $P(1)$ is true
- If $P(k)$, then $P(k + 1)$ is true

Why does this work? Does the sequence need to be finite?

Induction: Example

Example: Prove that the sum of the first n natural numbers is $\frac{1}{2}n(n+1)$.

■ **$P(1)$ is true:** Let $n = 1$. Then $\frac{1}{2} \cdot 1(1+1) = \sum_{j=1}^1 j = 1$.

■ **If $P(k)$, then $P(k+1)$ is true:**

- Now let $n = k$, and assume that $\sum_{j=1}^k j = \frac{1}{2}k(k+1)$.
- Add $k+1$ to both sides: $\sum_{j=1}^{k+1} j = \frac{1}{2}k(k+1) + k+1$
- $= \frac{1}{2}k^2 + \frac{1}{2}k + k + 1$
- $= \frac{1}{2}((k^2 + 2k + 1) + (k+1))$
- $= \frac{1}{2}(k+1)((k+1)+1)$.

Basic Properties of Arithmetic

- Associative: $(a + b) + c = a + (b + c)$
- Commutative: $a + b = b + a$
- Distributive: $a(b + c) = ab + ac$
- Identity: $x + 0 = x$ and $x * 1 = x$
- Inverse: $(-x) + x = 0$ and $x^{-1} * x = 1$ (Not a property of Integers)
- PEMDAS: Please Excuse My Dear Aunt Sally

Ratio, Proportion, and Percentages

- A **Ratio** of two quantities is one divided by the other: $\frac{x}{y}$ is the ratio of x to y . Also written $x : y$
- A **Proportion** of two variables is the amount one variable represents of itself and a second variable: $|\frac{x}{x+y}|$. The amount of a variable compared to the whole.
- Proportion over a range of 0 to 100
- **Percentage Change:** $\frac{x_{t+1} - x_t}{x_t}$

Factoring

- Consider the equation $\delta + \delta^2 + 4\delta - 6\delta^2 + 18\delta^3$. How do we simplify this equation?
- Consider the equation $3x + 4x^2$. How might we simplify this equation?
- Consider the polynomial $x^2 - 2x - 3$. How might we simplify this equation?
- Consider the fraction (with the polynomial in the numerator) $\frac{x^2-1}{x-1}$. How might we simplify this equation?

Expansion: The FOIL Method

How do expand the expression: $(x + 3)(5 - 7x)$?

- We use FOIL
 - Multiply the **F**irst terms
 - Multiply the **O**uter terms
 - Multiply the **I**nner terms
 - Multiply the **L**ast terms

Solving Quadratic Equations

Sometimes we will be confronted with a problem similar to the following: $(x + 5)^2 = 16$. How might we solve this problem?

- $x + 5 = 4$ or $x + 5 = -4$

- $x = -1$ or $x = -9$

Solving More Complex Quadratic Equations

Consider the quadratic equation $2x^2 + 16x - 2 = 0$

- Move the constant to the LHS of the equation $2x^2 + 16x = 2$
- Divide through by the coefficient on x^2 to get $x^2 + 8x = 1$
- Divide the coefficient on x by 2 and then square it to get 16
 - Add that value to both sides of the equation to get
$$x^2 + 8x + 16 = 17$$
- Factor the RHS of the equation to see if yields an $(x + \dots)^2$ or $(x - \dots)^2$
- Here we get $(x + 4)^2 = 17$
- Take the Square Root of both sides and simplify to get:
$$x = -4 + \sqrt{17} \text{ or } x = -4 - \sqrt{17}$$

General Solution for Solving Quadratic Equations

The general form of the quadratic equation is $ax^2 + bx + c = 0$.
The general solution to solve for x is:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Inequalities

If $x > y$, then

- $x + a > y + a$
- $ax > ay$ if $a > 0$
- $ax < ay$ if $a < 0$
- $\frac{x}{a} > \frac{y}{a}$ if $a > 0$
- $\frac{x}{a} < \frac{y}{a}$ if $a < 0$

Functions

A function takes an object (usually a number or an array of numbers) and assigns exactly one other object (usually another number or another array of numbers) to it. For example a utility function transforms consumption into utility: $U(p, b) = 2p + 3b$.

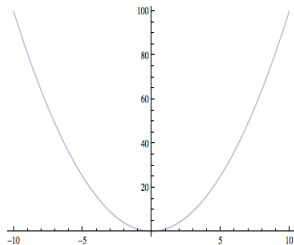
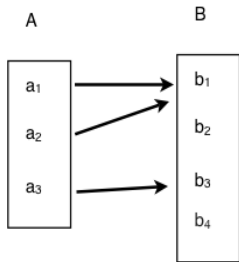
Definition

A *function* f from a set A to a set B is a rule that assigns to each element of A one and only one object in B . It is denoted $f : A \rightarrow B$. The set A is called the *domain* of f and the set B is called the *target space* or *codomain*, and $f(A)$ is called the *range* of f .

Functions can be **implicitly** defined $y = f(x)$ or **explicitly** defined $f(x) = x^2$.

Function Vs Correspondence

- A function assigns *exactly one* element of set B to an element of set A .
- If the rule assigns more than one element of set B to an element of set A , then the rule is called a correspondence.



$$f(x) = x^2$$

Graphs

Often we can use graphs to depict Functions and Correspondences. If the relation provides a mapping from the set of real numbers to the set of real numbers $\mathbb{R} \rightarrow \mathbb{R}$, then we can use a Cartesian or xy graph. For example we can graph the function $f(x) = x^2$ or the correspondence $x^2 + y^2 = 1$

Function Composition

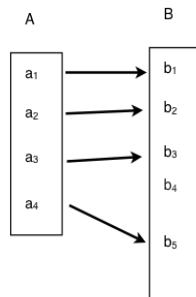
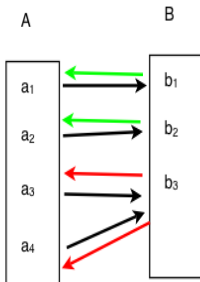
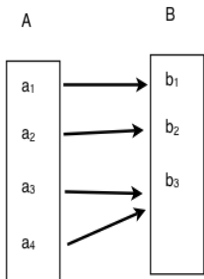
One can chain multiple functions, which is called a **Function Composition**. This is written as $g(f(x))$ or as $g \circ f(x)$.

- If we have $f(x)A \rightarrow B$ and $g(x)B \rightarrow C$, then $g \circ f(x)A \rightarrow C$
- For example, if we have $f(x) = 3x$ and $g(x) = x^2$, what is $g(f(x))$?
- Composition of functions is associative $f \circ (g \circ h) = (f \circ g) \circ h$
 - For example, $f(x) = 2x$, $g(x) = x^2$, and $h(x) = x + 2$. Then $g \circ h(x) = (x + 2)^2$ and $f \circ (g \circ h(x)) = 2(x + 2)^2$.
 - But it is also the case that $f \circ g(x) = 2x^2$ and $(f \circ g) \circ h(x) = 2(x + 2)^2$
- Composition of functions are not always commutative. It is not always the case that $f(g(x)) = g(f(x))$. For example, consider the case when $f(x) = 3x$ and $g(x) = x^2$.

Injective, Surjective, and Bijective

- The function f is called *injective* or *one-to-one* if $\forall b \in f(A), \exists$ only one $x \in A$ with $b = f(x)$. This means that every output of the function has at most one input. Or for any $x, y \in A$, if $f(x) = f(y)$, then $x = y$.
- f is called *surjective* or *onto* if for each element $b \in B \exists x \in A$ such that $b = f(x)$. Every element in B is assigned to an element in A . Or the $f(A) = B$
- f from A to B is called *bijective* if it is both onto and one-to-one.
- If a function is onto and one-to-one we can define the inverse function f^{-1}

Onto and One-to-One, Examples



Inverse Function

We can define an the *inverse function* f^{-1} from B to A such that $f^{-1}(f(x)) = x$ for all $x \in A$.

- If $f(x) = 4x + 10$, what is it's inverse?

- $f^{-1}(x) = \frac{x-10}{4}$

- If $f(x) = x^2$ and $A = \mathbb{R}$ can we define the inverse function?
What condition is violated?

- It is the case that $f^{-1}(f(x)) = f(f^{-1}(x))$

Functions in $\mathbb{R}$

Let X and Y be two subsets of $\mathbb{R}$ and $f : X \rightarrow Y$. Consider the following properties:

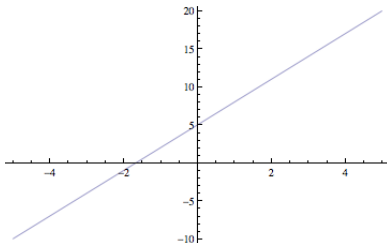
- **Monotonicity** Let x_1 and x_2 be any two numbers in X such that $x_1 < x_2$. f is monotonically increasing if $f(x_1) \leq f(x_2)$ and monotonically decreasing if $f(x_1) \geq f(x_2)$. f is *strictly* monotonically increasing (decreasing) if $f(x_1) < f(x_2)$ ($f(x_1) > f(x_2)$).
- **Boundedness** The function f is bounded if $\exists K \in \mathbb{R}$ such that $|f(x)| < K \ \forall x \in X$.

Functions in more than one Variable

Functions can provide a mapping from $A_1 \times A_2 \times \dots \times A_n \rightarrow B$. For example, $f(x, y) = x^2 + 3y$ or $f(x, y, z) = xy + 10z$. If the domain is two dimensions then the function can be graphed on a 3 dimensional plot.

Linear and Affine Functions

An affine function or a takes the shape of $y = a + bx$, where a and b are constants. This function is useful when the effects of x on y are constant. For example, $y = 5 + 3x$:



Technically this is not a linear function, which does not permit the constant, a . See the details on page 55-56.

Nonlinear Functions: Exponents

Exponents express a number multiplied by itself:

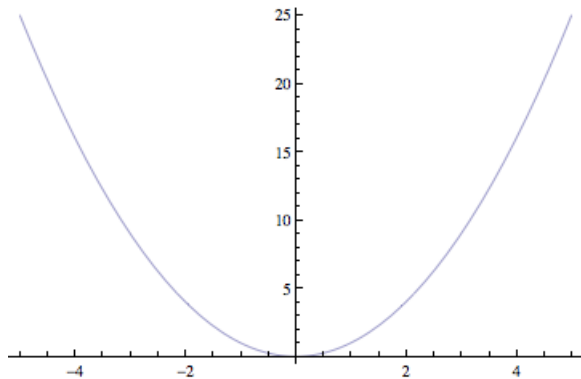
$x^n = x \times x \times \dots x$ (n times). We can also use exponent notation to denote the following relationships:

- $x^{-n} = \frac{1}{x^n}$
- $x^{\frac{1}{n}} = \sqrt[n]{x}$
- Solve 4^{-2}
- Solve $4^{\frac{1}{2}}$

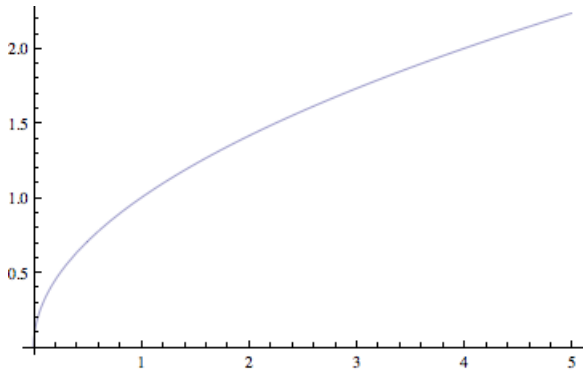
Properties of Exponents

- Multiplication: $x^m \times x^n = x^{m+n}$ and $x^m \times z^m = (xz)^m$
 - For example, $x^4 \times x^5 = (xxxx) \times (xxxxx) = xxxxxxxxx = x^9$
 - Consider $x^2 \times z^2 = (xx)(zz) = (xz)(xz) = (xz)^2$
 - Is it true that $x^3 \times z^5 = (xz)^7$?
- $(x^m)^n = x^{mn}$
 - For example, $(x^3)^2 = (xxx)(xxx) = x^6$
- $\frac{x^n}{x^m} = x^{n-m}$, since $x^{-m} = \frac{1}{x^m}$

Plotting $f(x) = x^2$



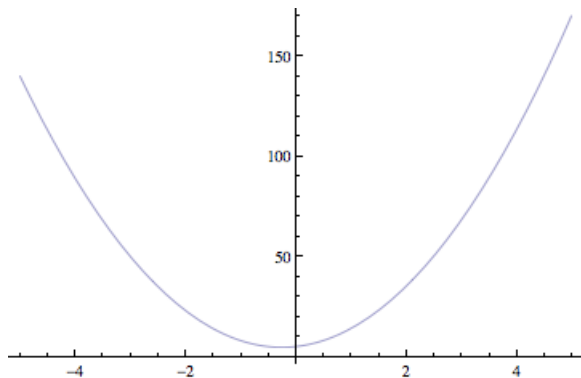
Plotting $f(x) = x^{\frac{1}{2}}$



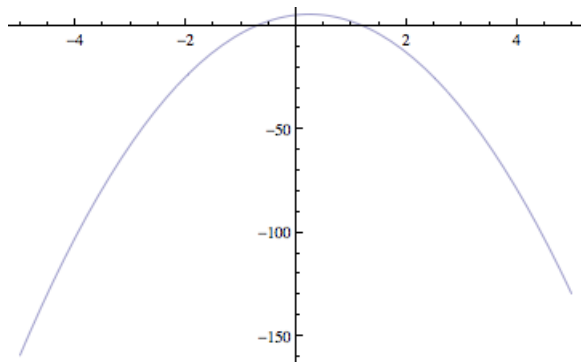
Quadratic Functions

If y is a **quadratic function** of x then $y = \alpha + \beta_1 x + \beta_2 x^2$. This is a non-linear function that describes a parabola. These functions allow one to depict a “U” relationship between variables. For example, protests might be less likely to arise in non-repressive settings and in highly repressive settings.

Plotting $y = 5 + 3x + 6x^2$



Plotting $y = 5 + 3x - 6x^2$

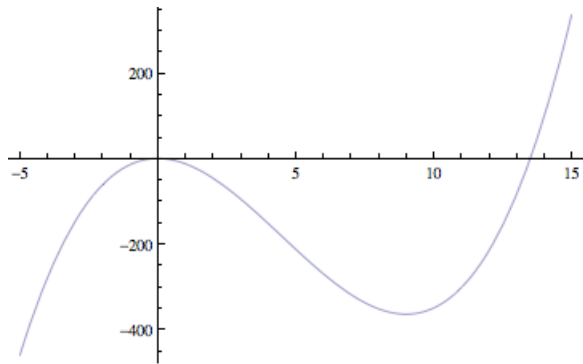


Higher-Order Polynomials

Polynomial Functions have the general form

$y = \alpha + \beta_1 x + \beta_2 x^2 + \dots + \beta_n x^n$, where $n < \infty$. These functions are not extremely common in applied political science work but are useful for later material. They allow one to depict relationships between variables when there are two or more thresholds over which the relationship between variables changes.

Plotting $y = x^3 - \frac{27}{2}x^2$



Logarithms

Logarithms can be understood as the inverse of exponents. Thus,

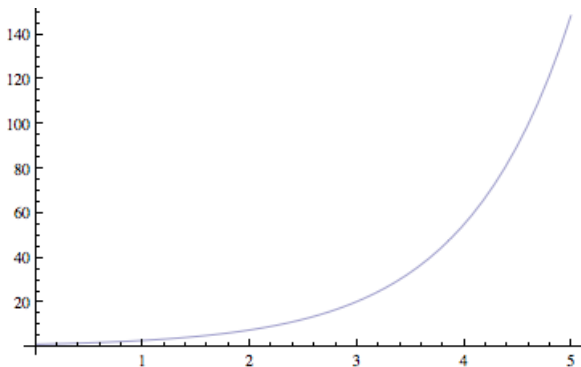
- $a^{\log_a x} = x$
- $\log_a a^x = x$
- If $\log_a x = b$, then $a^{\log_a x} = a^b$

Log and ln

The two most common bases for a logarithm are 10 and $e \approx 2.7183$. The function $f(x) = e^x$ has many important properties that we will use later. It is also used, among other reasons, to calculate compound interest and populations of animals. Often the log with a base of e is written as $\ln$, but it is also written as $\log$. For $\ln$, we have

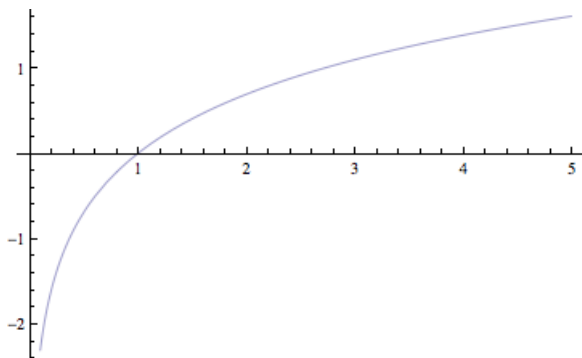
- $e^{\ln x} = x$ and $\ln(e^x) = x$

Plotting $f(x) = e^x$



This function is useful for depicting a relationship between two variables that increases at an increasing rate: Examples?.

Plotting $f(x) = \ln(x)$



This function is useful for depicting a relationship between two variables that increases at a decreasing rate: Examples?

Properties of Logarithms

- $\ln(x_1 \cdot x_2) = \ln(x_1) + \ln(x_2)$ and $\ln(\frac{x_1}{x_2}) = \ln x_1 - \ln x_2$
- $\ln(x_1 + x_2) \neq \ln(x_1) + \ln(x_2)$ and $\ln(x_1 - x_2) \neq \ln(x_1) - \ln(x_2)$
- $\ln(x^b) = b \ln(x)$
- Properties can be useful for transforming equations into simpler ones: Cobb Douglas Utility Function can be written as $U(p, b) = p^x b^{(1-x)}$ or the natural log of the utility function $u(p, b) = x \ln(p) + (1 - x) \ln(b)$

Absolute Value and Piecewise Functions

- $|x| = \sqrt{x^2}$
- Piecewise functions are functions defined in pieces. For example,

□

$$f(x) = \begin{cases} x + 5 & : x \leq 6 \\ x^2 & : x > 6 \end{cases}$$

- If is no chance of a democracy collapsing if GDP per capita exceeds \$6055, but that the likelihood increases as GDP per capita decreases below this level. Then we might write a function that defines the likelihood of collapse as a piecewise function.

Concavity and Convexity

- **Concavity:** The function f is *concave* if $\forall \alpha \in [0, 1]$ and $x_1, x_2 \in \mathbb{R}$, $f(\alpha x_1 + (1 - \alpha)x_2) \geq \alpha f(x_1) + (1 - \alpha)f(x_2)$
- **Convexity:** The function f is *convex* if $\forall \alpha \in [0, 1]$ and $x_1, x_2 \in \mathbb{R}$, $f(\alpha x_1 + (1 - \alpha)x_2) \leq \alpha f(x_1) + (1 - \alpha)f(x_2)$
- These are different concepts than a convex set

Preference Relations

We will consider three types of preference relations that a student can have between movies:

- **Strict Preference:** Student i can strictly prefer one movie to another. We use P_i to denote this preference relation.
- **Indifference:** Student i can be indifferent between two movies. We use I_i to denote this preference relation.
- **Weak Preference:** Student i can weakly prefer one one movie to another. We use R_i to denote this preference relation.

Property 1 of Rational Preferences: Completeness (or Comparability)

- Preference relations are complete if for any two alternatives (say, Z and L) it is the case that ZP_iL , LP_iZ , or ZI_iL .
- We can also say that two alternatives are comparable if ZR_iL , LR_iZ or both.
- What are some examples when this property might not pertain to an individual's preferences?

Property 2 of Rational Preferences: Transitivity

- The strict preference relation is transitive if ZP_iL and LP_iA , imply ZP_iA .
- The indifference relation is transitive ZI_iL and LI_iA imply ZI_iA . The weak preference relation is similarly defined.
- What are some examples when this property might not pertain to an individual's preferences?

Maximum Principle

- With completeness and transitivity an individual can order alternatives from the lowest to the highest alternative. A rational actor will select the highest alternative.

Utility Function

A **utility function** assigns a value, typically a number, from a set, say A , which depicts a relation. For example, if we assume people like money we could model the utility function as

- $u_i(x) = x$, where x is the amount of money individual i owns.
- Alternatively we could model the utility function as $u_i(x) = \ln(x)$.
- Alternatively we could model the utility function as $u_i(x)$ and give it properties such as $u_i(x)$ is concave and monotonically increasing.

Best Response Correspondences

Consider two states, s_1 and s_2 that can choose to invest in guns G or invest in domestic spending B . Say that s_1 is a hostile state that always prefers to invest in guns. Say that s_2 prefers to invest in guns if s_1 invests in guns, but prefers to invest in domestic spending if s_1 invests in domestic spending.

- For s_2 we can write a **best response function** that maps s_1 's action to s_2 most preferred action: $B_{s_2}(\cdot)$
 - $B_{s_2}(G) = ?$ and $B_{s_2}(B) = ?$
- Can we write a best response function for s_1 ?
- Yes, for s_1 , $B_{s_1}(G) = G$ and $B_{s_1}(B) = G$
- Could we write a best response for s_1 if s_1 is instead indifferent between G and B , when s_2 chooses B ?
 - Now we must write a **best response correspondence**, since $B_{s_1}(B) = \{G, B\}$